

1. The amount of federal tax per capita (per person) can be modeled by the function  $S = 3.4208t^3 - 55.88t^2 + 539.46t + 5118.8$  where  $t$  is the number of years after 1997.

- a. What are the values of  $t$  that correspond to 2008, 2018, and 2020?  
b.  $S = f(22)$  gives the value of  $S$  for what year? What is  $f(22)$ ?  
c. What  $x_{\min}$  and  $x_{\max}$  should be used to set a viewing window so that  $t$  represents 1997–2020?

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a. What value of  $t$  corresponds to the year 2008?

$t =$

What value of  $t$  corresponds to the year 2018?

$t =$

What value of  $t$  corresponds to the year 2020?

$t =$

b.  $S = f(22)$  gives the value of  $S$  for what year?

What is  $f(22)$ ?

$f(22) =$

(Round to the nearest integer as needed.)

c. What  $x_{\min}$  should be used for the viewing window?

$x_{\min} =$

What  $x_{\max}$  should be used for the viewing window?

$x_{\max} =$

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2. A company finds it can produce 20 heaters for \$5000, while producing 30 heaters costs \$7300. Express the cost,  $y$ , as a linear function of the number of heaters,  $x$ .

Express the cost,  $y$ , as a linear function of the number of heaters,  $x$ .

$y =$

(Type an expression using  $x$  as the variable. Simplify your answer. Do not factor.)

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3. As part of a damage award in a jury trial, a penalty handed down included a \$46 billion judgment against a company. If this amount was 92% of this company's revenue last year, how much was the company's revenue last year?

The company's revenue last year was \$  billion.

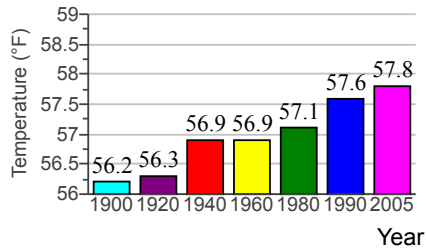
(Simplify your answer.)

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4. The total cost of a new automobile, including a 4% sales tax on the price of the automobile, is \$24,506. How much of the total cost of this new automobile is sales tax?

The sales tax is \$

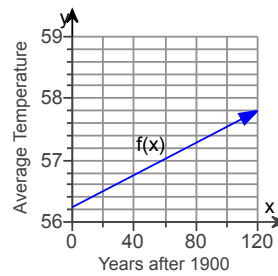
(Round to the nearest dollar as needed.)

- \*5. The bar graph shows the average temperature for a certain city, in degrees Fahrenheit, for seven selected years.



The data can be modeled by the linear function

$f(x) = 0.013x + 56.24$ , where  $f(x)$  is the average temperature, in degrees Fahrenheit,  $x$  years after 1900. The graph of  $f$  is shown to the right. Complete parts (a) through (b).



- a. Find and interpret  $f(20)$ . Use the equation for  $f(x)$  to identify this information as a point on the graph.

The point is .  
(Type an ordered pair.)

Interpret this information.

The average temperature in the city in the year  was  °F.

- b. Does  $f(20)$  overestimate or underestimate the actual data shown by the bar graph? By how much?

The value  $f(20)$  (1) \_\_\_\_\_ the data in the bar graph by  °F.

- (1) ☐ overestimates  
☐ underestimates

- \*6. Using a graphing calculator, estimate the real zeros, the relative maxima and minima, and the range of the polynomial function.

$$g(x) = x^3 - 1.2x + 1$$

The zero(s) of the function is/are at approximately  $x =$  .

(Round to three decimal places as needed. Use a comma to separate answers as needed.)

Find the relative maxima. Select the correct choice below and, if necessary, fill in the answer boxes to complete your choice.

- ☐ A. The function has a relative maximum of  at  $x \approx$  .  
(Round to three decimal places as needed.)

- ☐ B. The function does not have any relative maxima.

Find the relative minima. Select the correct choice below and, if necessary, fill in the answer boxes to complete your choice.

- ☐ A. The function has a relative minimum of  at  $x \approx$  .  
(Round to three decimal places as needed.)

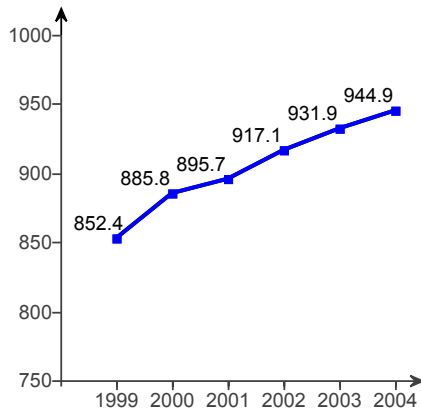
- ☐ B. The function does not have any relative minima.

The range of the function is .  
(Type your answer in interval notation.)

- \*7. After Mrs. Fisher lost 11% of her investment, she had \$31,150. What was Mrs. Fisher's original investment?

The original value of the investment was \$ .

- \*8. The graph below gives the number of millions of persons who took trips of 50 miles or more for the years 1999 through 2004.



Answer parts (a) through (c).

- (a) Find the equation of the line which is the best fit for these data, with  $x$  equal to the number of years after 1999 and  $y$  equal to the millions of persons.

$y =$

(Use integers or decimals for any numbers in the expression. Round to three decimal places as needed.)

- (b) Use the model to estimate the number of persons who took trips in 2008.

millions

(Round to two decimal places as needed.)

- (c) In what year does the model estimate the number of persons who took trips as 984.61 million?

(Round to the nearest whole number as needed.)

9. Suppose the monthly cost for the manufacture of golf balls is  $C(x) = 3200 + 0.42x$  dollars, where  $x$  is the number of golf balls produced each month. Also, suppose the monthly revenue for the sale of golf balls is  $R(x) = 1.22x$  dollars for  $x$  golfballs sold.

- (A) The model for profit, including output and input units, would be:

$P(x) =$

☐ A. dollars

for  $x$

☐ A. dollars

☐ B. golfballs

☐ B. golfballs

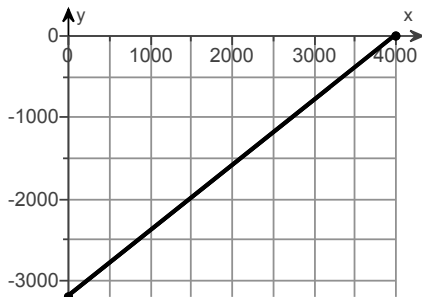
- (B) The rate of change for profit is

☐ A. dollars per

☐ A. dollar

☐ B. golfballs

☐ B. golfball



y - intercept

x - intercept

( 0, - 3200)

( 4000,0)

- (C) The y - intercept tells us:

- ☐ A. We have to sell 3200 golfballs to break even.  
☐ B. When we sell 0 golfballs we lose 3200 dollars.  
☐ C. When we sell 0 goldballs we get a 3200 dollar profit.

- (C) The x - intercept tells us:

- ☐ A. When we sell 0 goldballs we get a 4000 dollar profit.  
☐ B. We have to sell 4000 golfballs to break even.  
☐ C. When we sell 0 golfballs we lose 3200 dollars.

10. The total cost of a new automobile priced at \$24,038, including sales tax is \$25,480. Find the sales tax to the nearest percent.
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The sales tax rate is  %